

Divij Gera

SOFTWARE ENGINEER · MS COMPUTER SCIENCE · USC - VITERBI · GPA 4.0 / 4.0 · AVAILABLE MAY 2026
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EDUCATION

University of Southern California. Aug 2024 – May 2026
MS Computer Science · GPA 4.0 / 4.0 · Los Angeles, CA

Delhi Technological University. Aug 2019 – May 2023
B.Tech. Software Engineering · CGPA 9.09 / 10 · Delhi, India

Data Structures & Algorithms · Advanced Algorithms · Object Oriented Design · Operating Systems · Distributed Systems · Computer Networks · Machine Learning · Deep Learning · Natural Language Processing · Retrieval-Augmented Generation (RAG) · Information Retrieval

SKILLS

Languages
Cloud & Infra
Systems
AI & ML
Data & Web
Dev Tools

Python · Java · C++ · JavaScript / TypeScript · C# · .NET · Dart
AWS (EKS, Lambda, S3) · GCP · Azure · Kubernetes · Docker · GitHub · GitLab · CI/CD · Linux
Distributed Systems · TCP/IP · REST · Concurrency · Multithreading · Socket Programming
PyTorch · TensorFlow · Hugging Face · LLMs · Computer Vision · scikit-learn · InsightFace · LoRA fine-tuning · OpenAI API
Snowflake · SQL · ETL · Informatica · Pandas · Flask · Jupyter · Google Colab · Next.js · Firebase · Flutter · Unity (C#)
GitHub Copilot · Cursor · Claude · Gemini CLI

EXPERIENCE

Software Engineer
NatWest Group · Delhi, India

Jul 2023 – Jul 2024
AWS (EKS, Lambda, S3) · Docker · Kubernetes · GitLab · CI/CD

- Designed system architecture for migrating **70K+ repositories** from on-prem to AWS in an agile setup; containerized with Docker, orchestrated on Kubernetes (EKS) with horizontal auto-scaling, deployed with zero-downtime cutover for globally distributed teams
- Built GitLab CI/CD analytics on AWS, reducing pipeline failures by **~15%** and improving deployment times across distributed builds; cataloged **100K+ repositories** to enable code reuse across global teams

Software & Product Engineer
Get Me Therapy · Delhi, India

Jul 2022 – Jul 2023
LLM APIs · NLP · Sentiment Analysis · REST APIs · Cloud Deployment

- Prototyped NLP-driven features, including automated sentiment analysis of patient inputs and personalized therapy plan generation, serving 100+ daily sessions and reducing manual operational load by **40%**
- Translated clinical requirements into product specifications, drove end-to-end **MVP** development, and collaborated cross-functionally to ship an AI-enhanced therapy platform that contributed to **17% revenue growth**

Data Analytics Intern
NatWest Group · Delhi, India

Jun – Aug 2022
Snowflake · SQL · Informatica · Python · ETL Pipelines

- Migrated legacy ETL pipelines to Snowflake enabling analytics over **1B+ rows** of financial data; rewrote and optimized SQL workflows to unblock downstream predictive modelling integration while communicating technical decisions to stakeholders

PROJECTS

InstaNova: Cognitive Load Reduction Agent
Google DeepMind x InstaLILY Hackathon, San Francisco · github.com/geradivij/Instanova

Feb – March 2026
Python · FunctionGemma 270M · LoRA · MediaPipe · pytsx3

- Fine-tuned **FunctionGemma 270M** with LoRA adapters on a custom signal-to-action dataset; fuses webcam (eye strain + gesture), keyboard/mouse signals, and OS state into a real-time cognitive load score
- Built a full agentic **perceive → reason → act → observe → adapt** loop with persistent on-device memory; **zero cloud dependency** with all inference and user data running entirely on-device

Alzi: AI Assistant for Alzheimer's Patients
Cal Hacks 12.0 · San Francisco

October – November 2025
Python · Flask · InsightFace · LMNT · OpenAI APIs · Firebase · Flutter

- Integrated facial recognition (InsightFace), voice cloning (LMNT), and an LLM conversational agent into a Flask/Firebase backend achieving **<200ms** recognition latency
- Delivered an audio-first, on-device Flutter interface focused on accessibility for patients with cognitive decline

Bias Detection & Mitigation in Clinical LLMs
USC · github.com/geradivij/Bias-Detection-in-LLMs

Jan – May 2025
PyTorch · Hugging Face · Fairlearn · Pandas · ClinicalBERT · BioGPT · MIMIC-III

- Developed a bias detection framework using ClinicalBERT and BioGPT on MIMIC-III (**60K+ EHR records**) across mortality, length-of-stay, and treatment tasks; evaluated fairness across 4 demographic axes using 5 metrics, including Equalized Odds and Theil Index
- Applied ThresholdOptimizer, group-wise mean correction, and counterfactual augmentation to reduce disparities by **~50%** with zero accuracy degradation

Music Genre Transfer
USC · github.com/geradivij/Music-Genre-Transformation

Jan – May 2025
PyTorch · CycleGAN · Transformer · VAE · StarGAN · MusicGen · Flask · TypeScript

- Implemented and benchmarked **5 generative architectures** (CycleGAN, VAE, Transformer, StarGAN, MusicGen) on GTZAN (10 genres); MusicGen via prompt conditioning achieved best genre coherence across all models
- Built a real-time web demo enabling genre-consistent remixing and interactive listening evaluation

Pneumonia CNNs · github.com/geradivij/Pneumonia-CNNs

March 2021

- Built CNN-based pneumonia classification on chest X-ray images, benchmarking **27 variants** including VGGNet and transfer learning architectures - github.com/geradivij/Pneumonia-CNNs

PUBLICATIONS

- [ResearchGate](#): Large-scale sentiment analysis benchmarking Classical ML (SVMs, Logistic Regression, and Naïve Bayes) vs Deep Learning approaches (BERT Transformer and simple neural networks) on web datasets across accuracy and F1 score
- Feb 2022